

Charlotte Petersen

School of Chemistry
University of Melbourne
Melbourne, Australia
charlotte.petersen@unimelb.edu.au

Employment

November 2022 – Present

University of Melbourne, School of Chemistry

Lecturer and ARC DECRA Fellow

December 2021 – November 2022

University of Sydney, School of Chemistry

ARC DECRA Fellow

November 2020 – December 2021

University of Queensland, Australian Institute for Bioengineering and Nanotechnology.

Research Fellow in the Theoretical and Computational Molecular Science group of Debra Bernhardt.

September 2017 – October 2020

University of Innsbruck, Institute for Theoretical Physics, Austria

Lise Meitner Fellow in the Bio and Nano Physics group of Thomas Franosch

August 2015 – July 2017

Aalto University, Department of Applied Physics, Finland

Postdoctoral researcher in the Complex Systems and Materials group.

Supervisors: Prof. Mikko Alava and Prof. Stefano Zapperi

Education

2012 – 2016 Australian National University

Conferred December 2016

Doctor of Philosophy in Chemistry

Title: An Investigation Into the Significance of Dissipation in Statistical Mechanics.

Supervisors: Prof. Denis Evans and Dr. Stephen Williams

2008 – 2011 Australian National University

Bachelor of Philosophy (Science)

First class Honours in Chemistry, awarded the University Medal.

Honours thesis title: Computational Demonstration of the Fluctuation Theorem with a Parallel RC Circuit.

2010 University of California, Berkeley

One semester overseas exchange.

2006 – 2007 Narrabundah College

University Admissions Index (UAI) of 99.95, Dux of school

Research Fellowships

- 2021: Discovery Early Career Researcher Award (DECRA) (\$415,283)
Project title: Extracting the hidden structure of glass from particle vibrations.
- 2019: Austrian Science Fund (FWF) Lise Meitner Programme (EUR 156,140)
Project title: Characterization of liquids with modulated density profiles.

Funding, Scholarships, Prizes, and Awards

- 2025: ARC DP. Lead CI. Project title: Metallic glass nanomaterials: New theory and syntheses. (\$569,717)
- 2024: Women in Science Academic Award. (\$25,000)
Project title: Unlocking the potential of metallic glass through computer simulations
- 2024: Pawsey internship scheme, Primary Investigator, (\$12,600)
Project title: Machine learning interactions for metallic glass simulations
- 2024: National Computational Merit Allocation Scheme – Lead Chief Investigator (1000ksU $\approx$ \$40,000)
- 2023: Selby Research Award. (\$21,000)
Project title: Breaking the second law of thermodynamics with thermal fluctuations.
- 2023: PTEP Editors' Choice Award, The Physical Society of Japan
- 2023: National Computational Merit Allocation Scheme – Lead Chief Investigator (1000ksU $\approx$ \$40,000)
- 2022: Physical Chemistry Division ECR talk prize at the RACI Congress, Brisbane
- 2022: National Computational Merit Allocation Scheme – Lead Chief Investigator (500ksU $\approx$ \$20,000)
- 2021: Pawsey Centre for Extreme Scale Readiness (PaCER) program.
Project title: Towards a molecular level understanding of flow-induced physical and chemical reactions.
- 2020: Best ECR talk at the Statistical Mechanics of Soft Matter Meeting, Brisbane
- 2018: AIP 2018 ANN Education travel grant (\$373)
- 2017: Aalto School of Science Education Network in Condensed Matter and Materials Physics Travel Support (EUR 754)
- 2014: Australian Nanotechnology Network Student/ECR Travel Bursary (\$350)
- 2012: Australian Postgraduate Award (APA) (\$77,116)
- 2012: Alan Sargeson Merit Scholarship in Chemical Science (\$16,250)
- 2012: RSC Supplementary Scholarship (\$9,750)
- 2011: University Medal, ANU
- 2010: Endeavour University Mobility in Asia and Pacific Grant (\$2,500)
- 2008: ANU National Undergraduate Scholarship
- 2008: Australian National University Dux Award
- 2008: Lord Florey Student Prize (\$2,000)

Invited conference presentations

- Frontiers in Artificial Spin Ice 2025, Zurich, 11-13 June 2025 (declined due to maternity leave)
- 24th East Asia Workshop on Chemical Dynamics, Taiwan, 18-20 March 2024
- Molecular Modelling 2023 Conference, Wollongong, 7-10 December 2023
- Australian Materials Chemistry Conference, Fremantle, 30 November – 2 December 2023

- Workshop on Theory and Simulation of Nonequilibrium Fluids, Brisbane, 11 – 15 September 2023
- NSW Student Computational Chemistry Meeting, Kiama, 10-12 July 2023. Workshop presentation: Molecular dynamics simulations fundamentals
- Science Early Career Academic Network Research Summit, Melbourne, 24 March, 2023
- RACI Congress, Brisbane, 3 – 8 July 2022. Title: The Debye-Waller factor of amorphous glass
-Awarded early career researcher talk prize.
- Australian Institute of Physics Summer Meeting, Melbourne, 3 – 6 December 2019, title: Effects of quasi-confinement on a fluid
- Loch Lomond Workshop on Artificial Spin Ice, Glasgow, 26-28 June 2017, title: Investigating artificial spin ice through lattice design

Publications

D. Crater, A. C. Lätti, M. Saccone, K. Hofhuis, D. Miertschin, B. Regmi, B. Achinuq, A. Kleibert, **C. F. Petersen** and A. Farhan (2024) Ice-rule driven vertex frustration in a stretched pentagonal spin ice. *Physical Review Research*, 6, L042064.

C. F. Petersen, P. Harrowell (2023) Direct measurement of the structural change associated with amorphous solidification using static scattering of coherent radiation, *The Journal of Chemical Physics*, 159, 244506

S. Sanderson, **C. F. Petersen**, D. J. Searles (2023) Machine learning a time-local fluctuation theorem for nonequilibrium steady states, *Progress of Theoretical and Experimental Physics*, 8, 083A01 – **Received the Editors' Choice of PTEP award.**

M. A. Hunter, B. Demir, **C. F. Petersen**, D. J. Searles (2022) A new framework for computing a general local self-diffusion coefficient using statistical mechanics, *Journal of Chemical Theory and Computation*, 18, 6, 3357–3363

C. F. Petersen and D. J. Searles (2022) Equilibrium distribution functions: connection with microscopic dynamics, *Physical Chemistry Chemical Physics*, 24, 6383-6392

K. Hofhuis, **C. F. Petersen**, M. Saccone, S. Dhuey, A. Kleibert, S. van Dijken, and A. Farhan (2021) Geometrical frustration and competing orders in the dipolar trimerized triangular lattice, *Physical Review B*, 104, 014409, **-Highlighted as the editors' suggestion.**

L. S. Schrack, **C. F. Petersen**, M. Caraglio, G. Jung, T. Franosch (2021) Tagged-particle motion in quasi-confined colloidal hard-sphere liquids, *Journal of Statistical Mechanics: Theory and Experiment*, 043301

A. Farhan, M. Saccone, **C. F. Petersen**, S. Dhuey, K. Hofhuis, R. Mansell, R. V. Chopdekar, A. Scholl, T. Lippert, S. van Dijken (2020) Geometrical Frustration and Planar Triangular Antiferromagnetism in Quasi-Three-Dimensional Artificial Spin Architecture, *Physical Review Letters*, 125, 267203

G. Jung and **C. F. Petersen** (2020) Confinement-induced demixing and crystallization, *Physical Review Research*. 22, 033207

L. S. Schrack, **C. F. Petersen**, G. Jung, M. Caraglio, T. Franosch (2020) Dynamic properties of quasi-confined colloidal hard-sphere liquids near the glass transition, *Journal of Statistical Mechanics: Theory and Experiment*, 093301

C. F. Petersen, L. S. Schrack, T. Franosch (2019) Static properties of quasi-confined hard-sphere fluids, *Journal of Statistical Mechanics: Theory and Experiment*, 8, 083216

C. F. Petersen, T. Franosch (2019) Anomalous transport in the soft-sphere Lorentz model, *Soft Matter*, 15, 3906

A. Farhan, M. Saccone, **C. F. Petersen**, S. Dhuey, R. V. Chopdekar, Y.-L. Huang, N. Kent, Z. Chen, M. J. Alava, T. Lippert, A. Scholl, S. van Dijken (2019) Emergent magnetic monopole dynamics in macroscopically degenerate artificial spin ice, *Science Advances*, 5, eaav6380

M. Hanifour, **C. F. Petersen**, M. J. Alava, S. Zapperi (2018) Mechanics of disordered auxetic metamaterials, *The European Physical Journal B*, 91, 271

C. F. Petersen, A. Farhan, S. Dhuey, Z. Chen, J. M. Alava, A. Scholl, S. van Dijken (2018), Tuning magnetic ordering in a dipolar square-kite tessellation, *Applied Physics Letters*, 112, 092403

A. Farhan, **C. F. Petersen**, S. Dhuey, L. Anghinolfi, Q. H. Qin, M. Saccone, S. Velten, C. Wuth, S. Gliga, P. Mellado, M. Alava, A. Scholl, S. van Dijken (2017) Nanoscale Control of Competing Interactions and Geometrical Frustration in a Dipolar Trident Lattice. *Nature Communications*, 8, 995

A. Farhan, A. Scholl, **C. F. Petersen**, L. Anghinolfi, C. Wuth, S. Dhuey, R. V. Chopdekar, P. Mellado, M. J. Alava, S. van Dijken (2016) Thermodynamics of emergent magnetic charge screening in artificial spin ice. *Nature Communications*, 7, 12635

C. F. Petersen, D. J. Evans, S. R. Williams (2016) Dissipation in monotonic and non-monotonic relaxation to equilibrium. *The Journal of Chemical Physics*, 144, 074107

C. F. Petersen, D. J. Evans, S. R. Williams (2016) Mechanism for asymmetric bias in demonstrations of the NPI and fluctuation theorem. *Molecular Simulation*, 42, 531-541.

C. F. Petersen, E. Krausz, D. J. Evans, S. R. Williams (2014) Theoretical Analysis of the Fluctuation Theorem Applied to Electric Circuits. *Communications in Theoretical Physics*, 62, 476-484

C. F. Petersen, D. J. Evans, S. R. Williams (2013) The instantaneous fluctuation theorem. *The Journal of Chemical Physics*, 139, 184106

M. Bulbrook, M. Chu, K. Deane, R. J. Doyle, J. Hinc, **C. F. Petersen**, G. Salem, N. Thorman, A. C. Willis (2010) Chiral Birch reduced tertiary phosphines: precursors to asymmetric 1,2-cyclohexenebis (tertiary phosphines). *Dalton Trans*, 39, 8878-8881.

Student Supervision

PhD students:

2024: Mathushan Sathiyamoorthy. Research topic: Computational studies of amorphous gold nanoparticles.

2021: Co-supervised Michelle Hunter. Research topic: Computational Studies of Transport Properties in Nanomaterials

Master's thesis students:

2023: Anna Findlay, Research topic: Computer simulations of polymers to measure structure

2024: Liang Tan, Research topic: Violations of the second law of thermodynamics in nanomagnet arrays

Bachelor's thesis projects:

2020: Beatriz Barroso Gstrein. Thesis title: Statistical physics of cluster formation in correlated nanomagnets

2019: Leah Rank. Thesis title: Movement of emergent magnetic monopoles in lattices of nanomagnets

2018: Ben Silwa. Thesis title: Disorder in Models of Artificial Spin Ice.

Undergraduate project students:

2025: Lisa Ye. Topic: Gold Nanoparticles in Cancer Treatment (Amgen Scholar)

2025: Aidan Dengler & Prasanth Devireddy. Topic: Machine learning interactions for metallic glass simulations (Pawsey Interns)

2024: Chloe Wang. Topic: Short-range interactions in a thermally active artificial spin ice

2024: Anya Latti. Topic: Towards Gold Glass Nanoparticles: Simulated Cooling of Gold Alloys to form Metallic Glasses

2022: Jim Mussared & Joseph McCormik. Topic: Unlocking the mysteries of glass.

Teaching

2023 - 2024 The University of Melbourne

- Reactivity and Mechanism – 3rd year physical chemistry (lecturer)
- Specialised topics in chemistry – statistical mechanics (lecturer)
- Environmental Chemistry (lecturer)
- Advanced Practical Chemistry (senior Demonstrator)
- Chemistry 1 (tutor)

2022 The University of Sydney

- Taught scientific communication skills course for HDR students.

2021 The University of Queensland

- Computational chemistry labs for Advanced Physical Chemistry (tutor)
- Chemistry 2 (teaching assistant)
- Assessed student presentations for the Physical Chemistry Honours course.

2017 – 2019 University of Innsbruck

- Practical molecular dynamics section of the Computational Physics course (tutor, course designer).
- Classical Mechanics (tutorials)
- Electrodynamics (tutorials)
- Fluid Mechanics (tutorials)

2011 - 2012 Australian National University

- Chemistry 1 (laboratory demonstrator)
- Chemical Structure and Reactivity 1 (laboratory demonstrator)
- Chemical Structure and Reactivity 2 (laboratory demonstrator)

Outreach and Engagement

2024: Panelist at the STEM EDGE program for year 7 students to improve gender equity in STEM

2024: Presented “Day in the life of a theoretical chemist” talk to high school students.

2023: Presented research work to undergraduate students

2023: Engaged with prospective students at University of Melbourne Open day

2023 & 2024: Secretary of Melbourne University of Chemistry Society

2022: Glass expert for ANSTO’s National Science Week Hackathon competition for school students.

2021: Member of the AIBN Seminar Committee.

2019: Organised physics activities for the public at the University of Innsbruck Festival of Science.

2019: Organised the Bio and Nano Physics station for the University of Innsbruck Public Open Day.

Contributed Conference Presentations

- Statistical Mechanics of Soft Matter, Sydney, 25-27 November 2024
- RACI Physical Chemistry Division Conference, Sydney, July 2024
- Statistical Mechanics of Soft Matter, Perth, 20-23 November 2023
- Statphys28, Tokyo, 7-11 August, 2023
- 9th International Discussion Meeting on Relaxations in Complex Systems, Chiba, 12-18 August, 2023
- XVI International Workshop on Complex Systems, Andalo, 13 -17 March 2023
- RACI Physical Chemistry Summer Festival, virtual, November 2021
- Statistical Mechanics of Soft Matter, Brisbane, 14-15 December 2020
-*Awarded best early career researcher talk prize.*
- Statistical Mechanics of Soft Matter, Adelaide, 16-17 December 2019
- DPG Spring Meeting of the Condensed Matter Section. Regensburg, 1– 5 April 2019
- XV International Workshop on Complex Systems, Andalo, 17 -20 March 2019
- 23rd Australian Institute of Physics Congress, Perth, 9-13 December 2018
- Statistical Mechanics of Soft Matter, Auckland, 6-7 December 2018
- DPG Spring Meeting of the Condensed Matter Section. Berlin, 11 - 16 March 2018
- Multiscale Material Modeling, Dijon, 9-14 October 2016

- Molecular Modeling 2014, Students and Early-Career Researchers' Forum, Queensland, 30-31 July 2014
- Statistical Mechanics of Soft Matter, Melbourne, 21-22 November 2013

Poster Presentations:

- Interdisciplinary Challenges in Nonequilibrium Physics, Vienna (Online), 12 – 16 April 2021
- Beg Rohu Summer School 2019: Glasses, Jamming, and Slow Dynamics, Saint-Pierre-Quiberon, 23 June – 7 July 2019
- International Soft Matter Conference, Edinburgh, 3-7 June 2019
- Ian Snook Conference on Chemical Physics, Melbourne, 4-5 December 2014
- Molecular Modeling 2014: From Biomolecules to Materials, Queensland, 31 July – 2 August 2014